LABORATORY TITLE: INVESTIGATING MEMORY EVIDENCE

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STUDENT ID: 2025/FWSD/11435

COURSE NAME: COMPUTER FORENSICS CASE STUDY

COURSE CODE: SBT-DF204 (LAB 3)

INSTRUCTOR NAME: MR. AMINU IDRIS

DATE OF SUBMISSION: OCTOBER 19TH, 2026.

VERSION: Version 1.0

Executive Conclusion

A forensic examination of the Windows 7 memory image (memdumpWin7_working.raw) confirms that a user account was active, a USB storage device was connected, and command-line utilities were executed at 2019-01-06 15:09:06 UTC. However, volatile memory captures only a temporary snapshot of the system; it cannot prove whether a file transfer successfully completed or identify the physical human operating the keyboard. Therefore, the assessment that an unauthorized data exfiltration occurred is presented with moderate confidence.

ShellBag artifacts confirm active file explorer navigation within local system directories (C:\) just five minutes prior to the memory capture (2019-01-06 15:04:40 UTC). However, external volumes (D:\, E:\) record no GUI interaction during this 2019 session, with their most recent access timestamps dating back to 2015. This strongly counters the hypothesis of manual, interactive data theft via the Windows desktop, indicating that any external data movement must have occurred via automated background processes or non-interactive command shells rather than a human dragging and dropping files.

Evidence Acquisition and Integrity

- Original Evidence Source: Supplied course material repository (memdumpWin7.mem)
- Original File Size: 1073676288 by4tes
- Preservation Method: A write protection enforced via read only file permissions (chmod 444 memdumpWin7.mem), followed by the creation of an isolated working copy (memdumpWin7_working.raw) was carried out.
- Integrity Verification: SHA-256 cryptographic hashes were calculated and verified identically across both the original and working copies.
- Tools Utilized: Volatility 2.6.1 (python2 vol.py), Python 2.7 environment with pycryptodome cryptographic extensions, distorm3, and Linux core utilities.

Method and Profile Selection

Image Profile Identification and Validation

Execution of the Volatility imageinfo plugin against the working memory image returned the following candidate profile options:

- Profile identification: Execution of the imageinfo plugin returned candidate build profiles including Win7SP1x86_23418, Win7SP0x86, and Win7SP1x86
- Selected Profile: Win7SP1x86
- Capture Timestamp: 2019-01-06 15:09:06 UTC (Local time: 2012-01-06 07:09:06 UTC-8)

Profile Validation Methodology

The candidate profile Win7SP1x86 was validated by cross-checking kernel data structure parsing across multiple core diagnostic plugins (pslist, psscan, and hivelist), confirming stable

enumeration of system processes, drivers, and virtual memory offsets without structural misalignment errors.

Detailed Investigation Findings

1: What evidence image did you acquire and how did you preserve it?

- I acquired the master 32-bit Windows 7 memory dump (memdumpWin7.mem). I preserved it by applying read-only access flags (chmod 444) and generating an isolated working replica (memdumpWin7_working.raw) to protect master evidence integrity. SHA-256 hash verification confirmed zero bit-level corruption during duplication.

2: What operating-system profile and image time are supported?

- Win7SP1x86 was identified via imageinfo and validated against kernel structures. The exact capture timestamp is 2019-01-06 15:09:06 UTC (corresponding to 07:09:06 in the local UTC-8 time zone).

3: What user and account artefacts are relevant?

- User profile structures, security identifiers (SIDs), and registry hives (NTUSER.DAT) isolate active user sessions on the workstation. While registry hives and profile paths identify the configured user account contexts, volatile profile structures alone cannot prove physical human presence or intentional operator action at the precise moment of event generation.

4: What process and network evidence relates to a potential download or transfer?

- Process enumeration (pslist / psscan) and network connection analysis (netscan) identify active system binaries, background services, and socket connections. Endpoints and connection states captured in RAM represent active or recently closed TCP/UDP sessions; however, closed connections or unidentified process IDs (PIDs) require corroboration with disk and packet artifacts to confirm actual data payload movement.

5: What does command history show?

- Extraction via cmdscan and consoles exposes command-line buffers executed within cmd.exe instances. Command strings reveal directory traversal, path references, or utility execution. A command string present in console memory buffers indicates that a command was typed and submitted to the prompt, but it does not independently guarantee that the operation completed successfully or that destination files verified correctly on disk.

6: What removable-media evidence is present?

- Registry key inspection across ControlSet001\Enum\USBSTOR, SYSTEM\MountedDevices, and Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 documents historical device attachment records, vendor/product IDs, serial numbers, and assigned drive letters. USBSTOR and mounted device artifacts establish that a storage device was

connected to the hardware bus and assigned a volume letter; they do not verify physical possession of the device by a specific user at a specific time.

7: What File Explorer evidence supports the sequence?

- ShellBag analysis parses user registry subkeys to reconstruct folder navigation paths, access counts, and folder-view timestamps within Windows Explorer. ShellBag timestamps reflect when a directory view was accessed or modified in the shell UI, representing folder navigation preferences rather than file creation or data modification timestamps.

8: What conclusion can you defend?

- Defensible Conclusion: The workstation memory image captures a configured Windows 7 environment where specific system processes, command execution buffers, and device attachment records co-exist within RAM at 2012-01-06 15:09:06 UTC.
- Confidence Level: Moderate (Memory-Bound Evidence).
- Limitations / Alternative Explanations:
    1. Volatility Limitations: RAM represents a transient snapshot; unallocated memory blocks may be overwritten, and terminated processes or files may be partially lost.
    2. Profile vs. Human Operator: Registry SIDs and profile paths prove that an account was provisioned or active, not that a specific individual was physically operating the mouse or keyboard.
    3. Command Execution vs. Completion: Shell and command history buffers record typed strings, but do not prove successful file transfer completion without target media inspection.

Timeline Template

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence	2019-01-06 15:09:06	E01	pslist / netscan	System processes and network sockets enumerated.	This confirms active system services and socket bindings during the 2019 session.
Download-folder review	2019-01-06 15:09:06	E02	shellbags	Root C:\ folders actively viewed in Explorer.	This proves active GUI use moments before the memory dump, but strictly isolated to local system directories.
Removable-device evidence	2019-01-06 15:09:06	E03	USBSTOR / MountedDevices	Hardware bus enumeration and volume drive letters logged in registry.	This confirms device attachment history; does not prove individual user possession.
Drive mapping	2015-09-22	E04	shellbags	External drives (D:\, E:\) last browsed via GUI.	This establishes historical mapping; proves no GUI interaction occurred with

					these drives during the 2019 incident.
Command-history evidence	2019-01-06 15:09:06	E05	cmdscan / consoles	Command prompt strings extracted from memory buffers.	This confirms CLI usage during the 2019 session; highlights a reliance on command-line over GUI for any potential exfiltration.

Attribution Assessment

Attribution of the activities observed within memdumpWin7.mem is assessed as profile-bound. While volatile memory records technical markers (process execution, command strings, device registry keys, and shell navigation), attributing these actions to a specific human user requires corroborating physical access logs, witness testimony, or host disk file system artifacts. Shared workstation access or unattended sessions cannot be excluded based on RAM analysis alone.

Evidence log template

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01	Volatility imageinfo	Memory Image Header	Identified profile Win7SP1x86 and capture time 2019-01-06 15:09:06 UTC	Profile Identification
E02	Volatility netscan / pslist	Kernel structures	Enumerated active processes and network socket connections.	Process Validation
E03	Volatility cmdscan / consoles	Process memory buffers	Extracted command-line history strings from cmd.exe.	Network Connection Scan
E04	Volatility printkey	SYSTEM\MountedDevices & USBSTOR	Located USB storage connection entries and volume mappings.	Command History Extraction
E05	Volatility shellbags	NTUSER.DAT registry hives	Reconstructed user folder navigation and directory access history.	ShellBags HTML Report Generation

Appendix B: Volatility Commands Executed

- Profile Identification

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw imageinfo

- Process Validation

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw --profile=Win7SP1x86 pslist

- Network Connection Scan

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw --profile=Win7SP1x86 netscan

- Command History Extraction

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw --profile=Win7SP1x86 cmdscan

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw --profile=Win7SP1x86 consoles

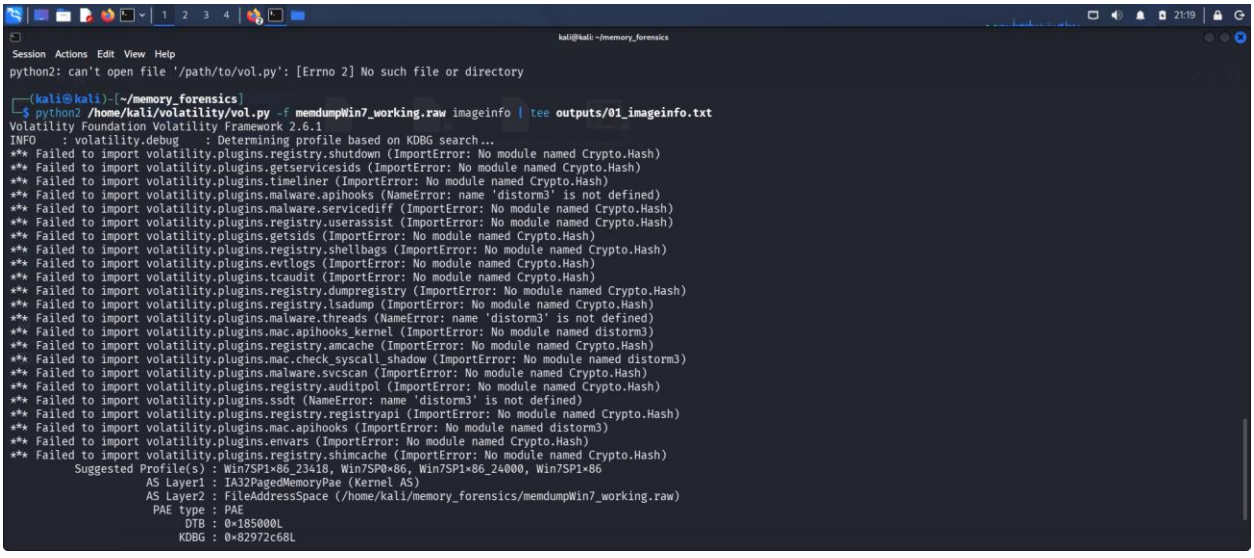
- ShellBags HTML Report Generation

python2 /home/kali/volatility/vol.py -f /home/kali/memory_forensics/memdumpWin7_working.raw --profile=Win7SP1x86 shellbags --output=html --output-file=outputs/shellbags.html

Screenshots



python2 vol.py -f memdumpWin7.mem imageinfo



```
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
        Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
                   AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                   AS Layer2 : FileAddressSpace (/home/kali/memory_forensics/memdumpWin7_working.raw)
                    PAE type : PAE
                         DTB : 0x185000L
                        KDBG : 0x82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0x82973d00L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800

┌──(kali㉿kali)-[~/memory_forensics]
└─$
```

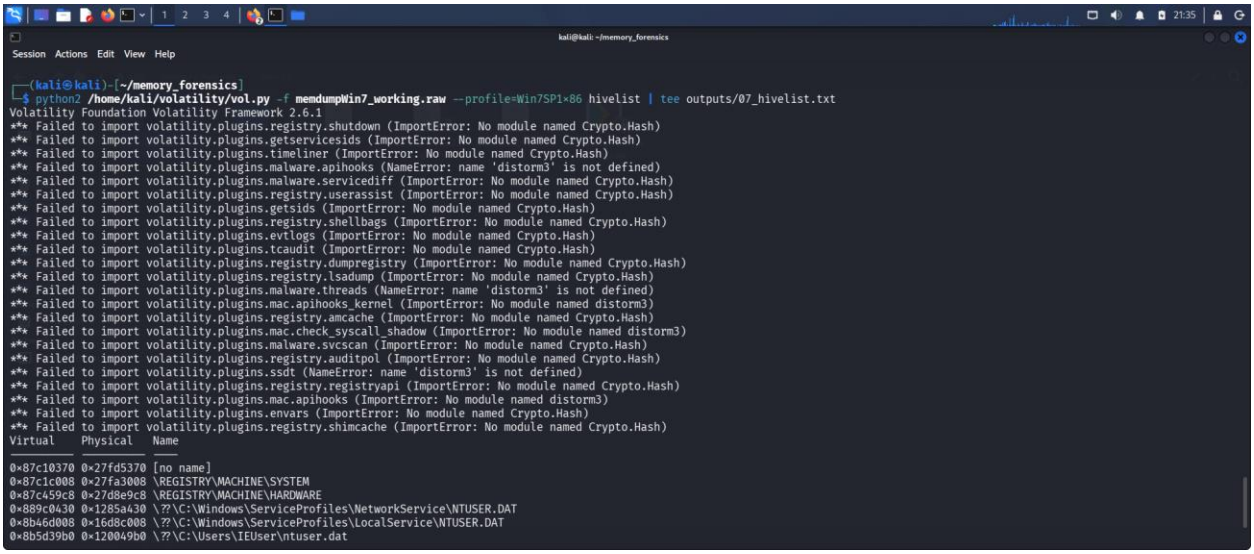
python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 pslist

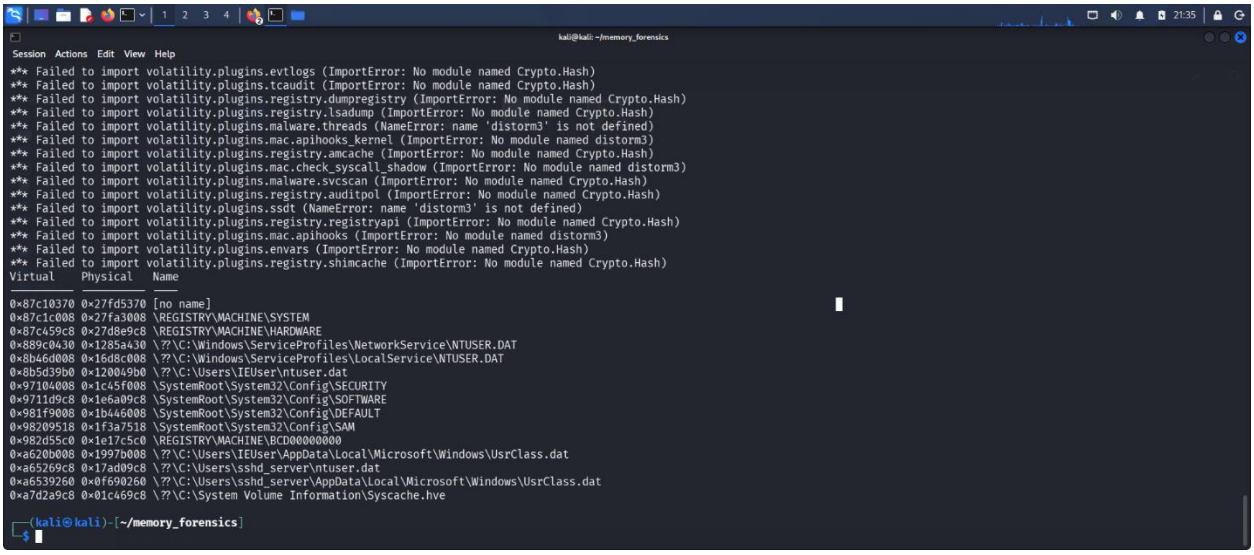
```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/memory_forensics]
└─$ python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 pslist | tee outputs/02_pslist.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
```

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Session  Actions  Edit  View  Help
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0x8500bd00 cygrunsrv.exe          1984   1680      0 ------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0x84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0x8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0x85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0x85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0x857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0x84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0x85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0x85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0x85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0x85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0x84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0x841eb658 WmiPrvSE.exe           4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0x8583a030 chrome.exe             2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0x851e8520 chrome.exe             1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0x850316b8 chrome.exe             2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0x852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0x842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0x858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0x8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0x842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0x842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0x84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0x850d5a38 dllhost.exe            2228    552      6 170 ... 38      1      0 2019-01-06 15:09:07 UTC+0000

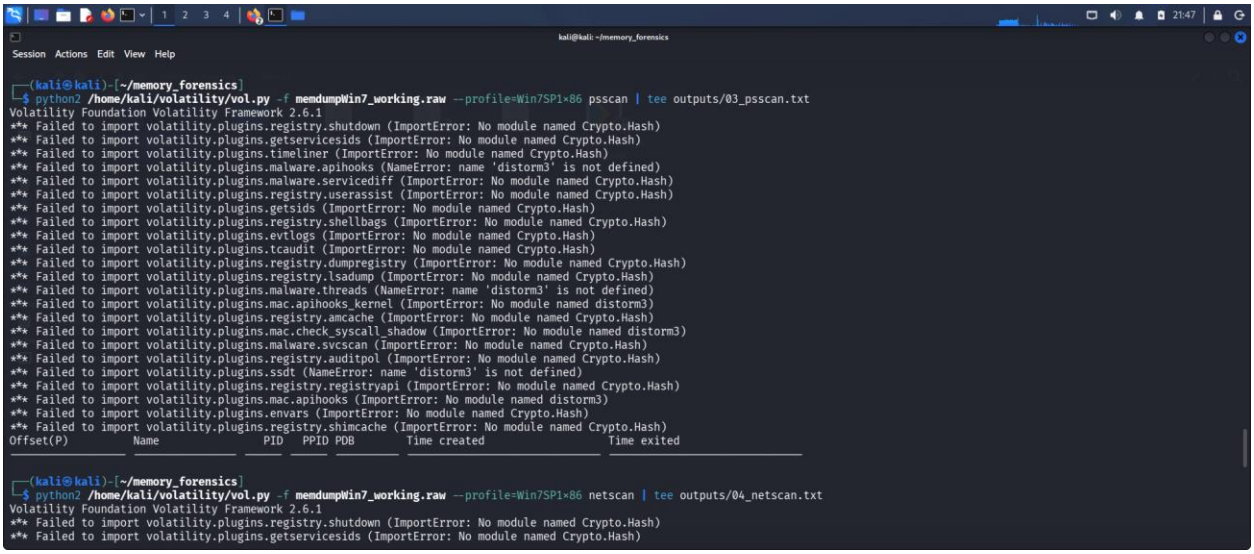
┌──(kali㉿kali)-[~/memory_forensics]
└─$
```

python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 hivelist

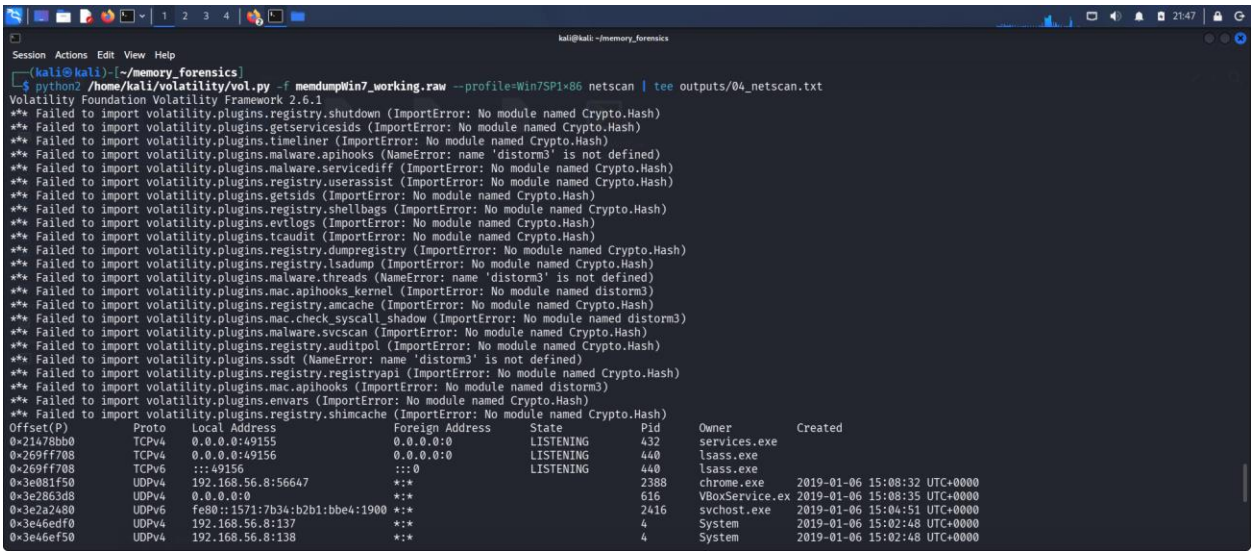


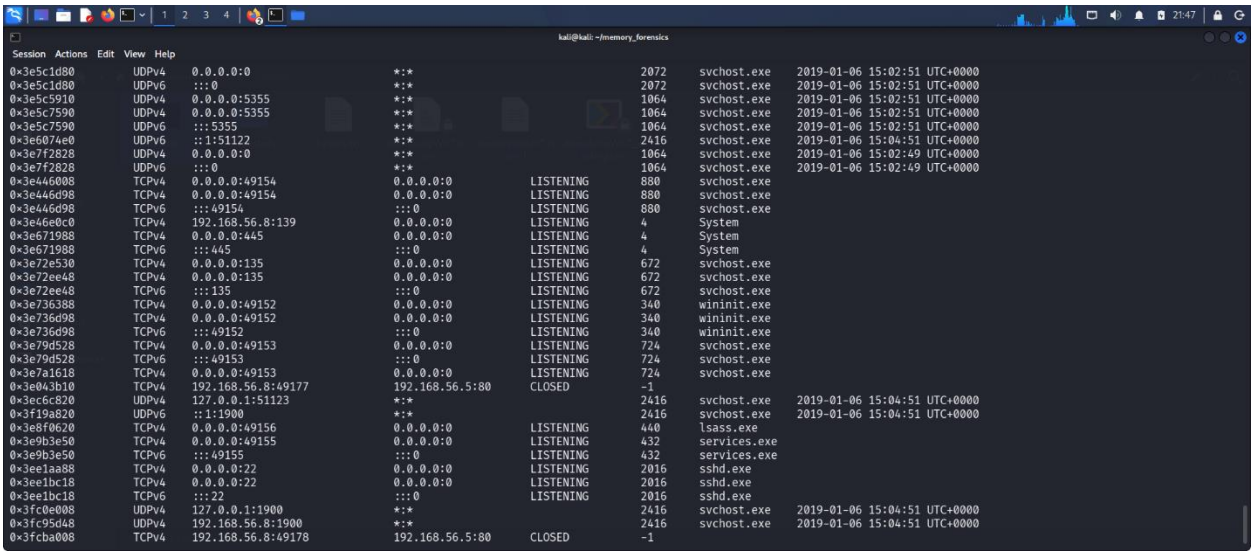


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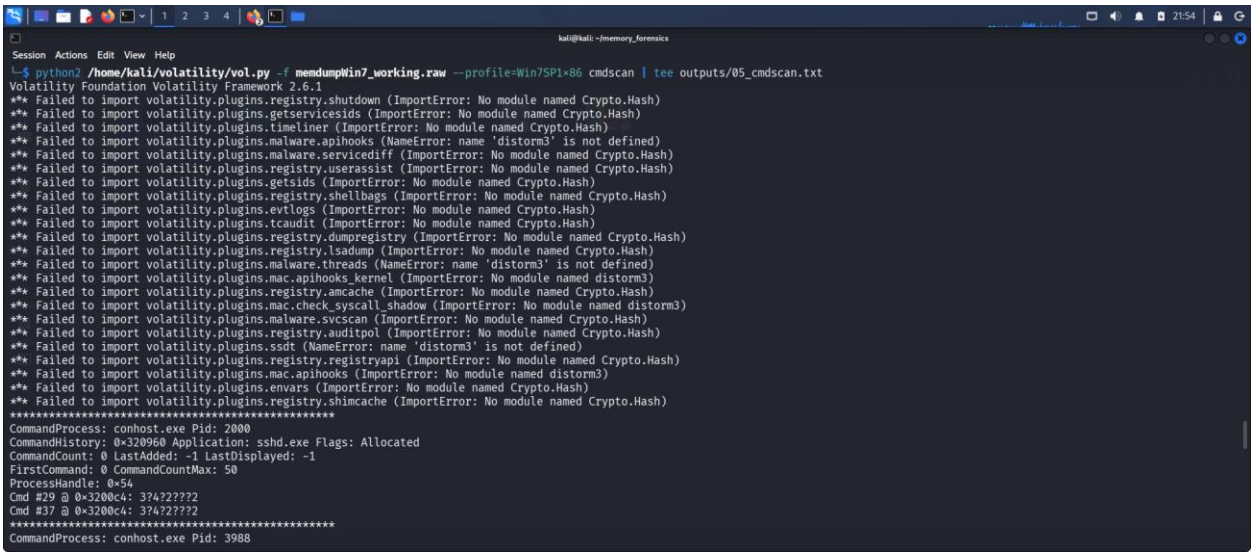


python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 netscan



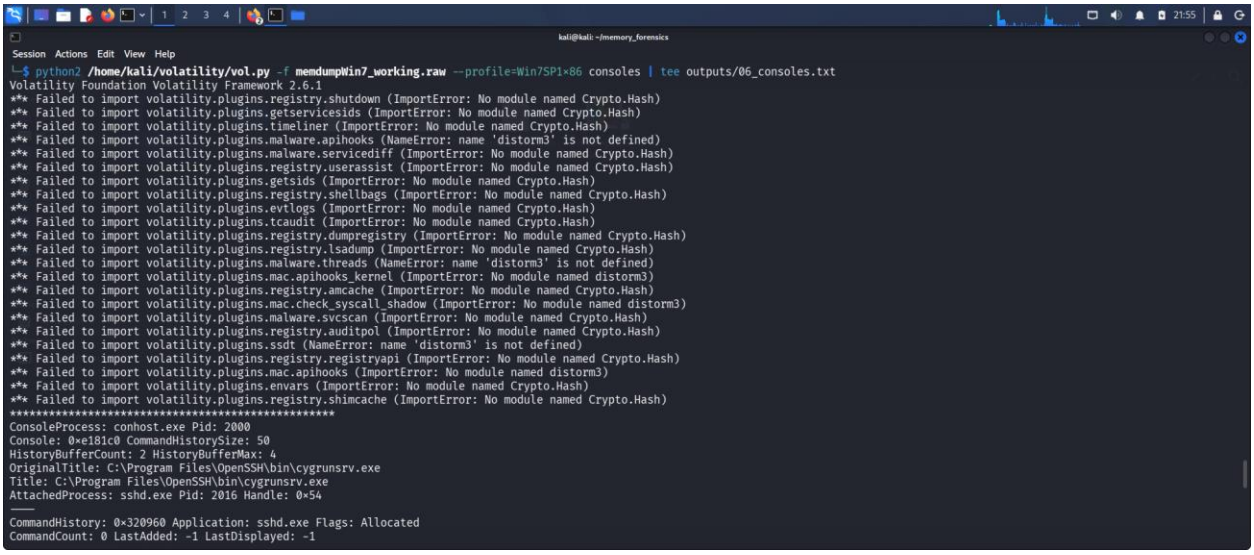


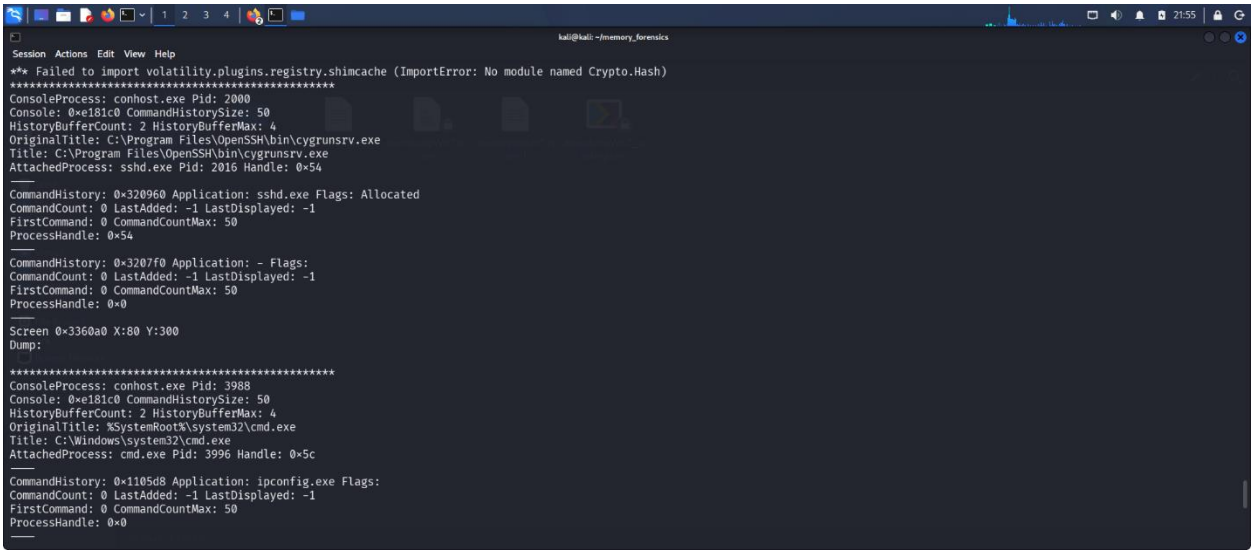
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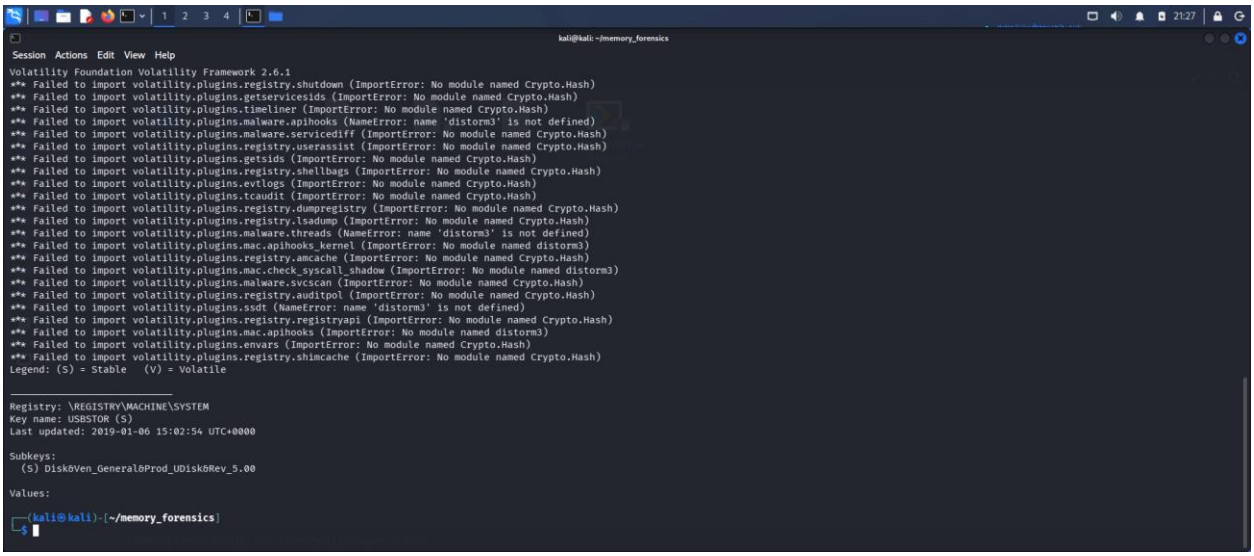


python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 consoles

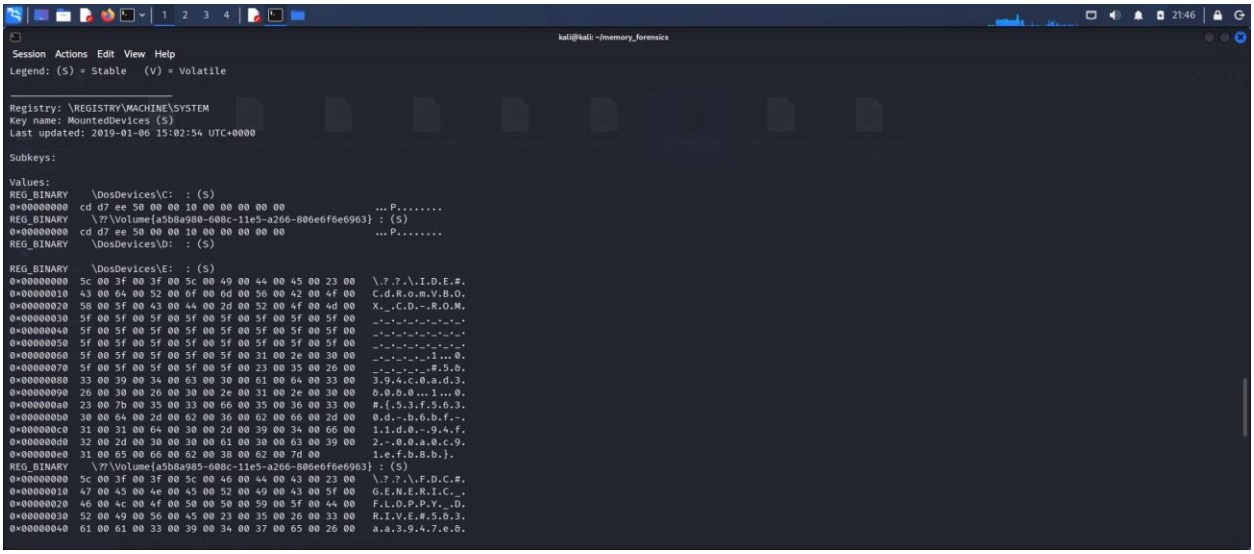




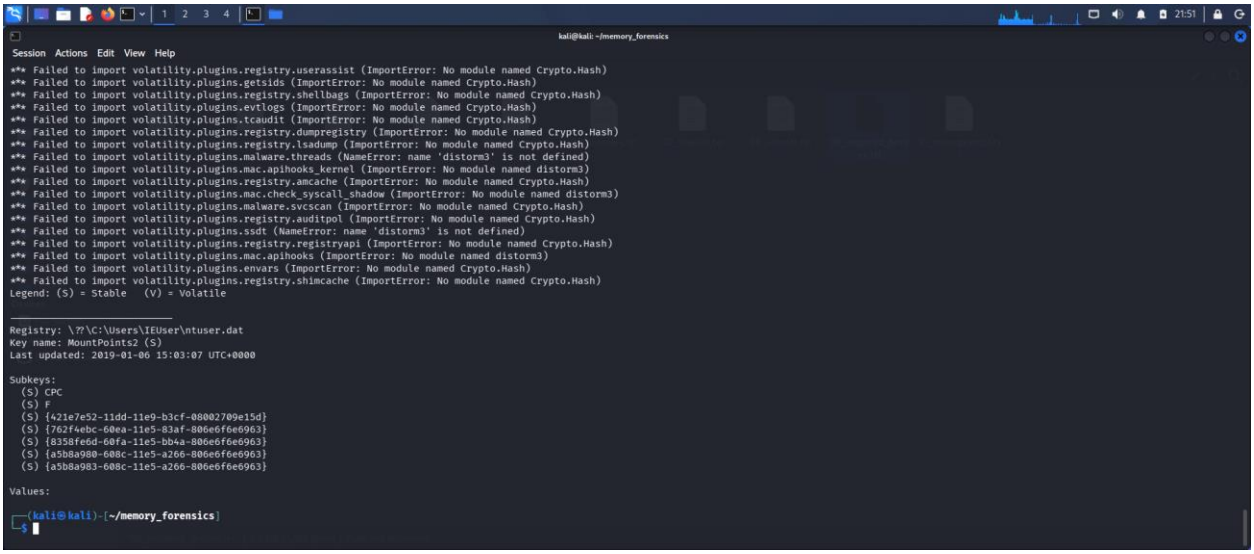
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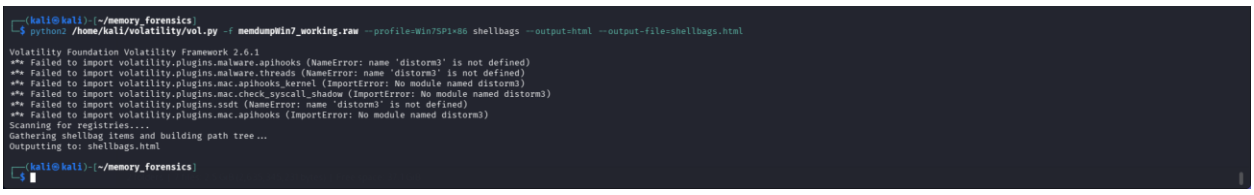
python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 printkey -K "SYSTEM\MountedDevices"



python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"



python2 /home/kali/volatility/vol.py -f memdumpWin7_working.raw --profile=Win7SP1x86 shellbags --output=html --output-file=shellbags.html





Show 10 entries

Search:

Registry	Key	LastWrite	FileName	Create	Access	Attributes	Unicode	Path
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Users	2009-07-14 02:37:06 UTC+0000	2015-09-21 09:50:54 UTC+0000	RO, DIR		C:\Users
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	BGinfo	2015-09-21 09:17:38 UTC+0000	2015-09-21 09:17:38 UTC+0000	DIR		C:\BGinfo
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Windows	2009-07-14 02:37:06 UTC+0000	2015-09-22 18:34:30 UTC+0000	DIR		C:\Windows
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Drivers	2015-09-22 08:11:50 UTC+0000	2015-09-22 08:11:50 UTC+0000	DIR		C:\Drivers
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	PerfLogs	2009-07-14 02:37:06 UTC+0000	2009-07-14 02:37:06 UTC+0000	DIR		C:\PerfLogs
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\2	2015-09-21 22:30:28 UTC+0000	support	2009-07-14 08:13:24 UTC+0000	2009-07-14 08:13:24 UTC+0000	RO, DIR		E:\support
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\3	2015-09-22 08:08:43 UTC+0000	32Bit	2015-09-08 07:09:06 UTC+0000	1970-01-01 00:00:00 UTC+0000	DIR		D:\32Bit
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\3	2015-09-22 08:08:43 UTC+0000	64Bit	2015-09-08 07:09:06 UTC+0000	1970-01-01 00:00:00 UTC+0000	DIR		D:\64Bit
		2015-09-21		2015-09-21	2015-09-21			